Lines and Angles

The big idea

Angles are everywhere: roof trusses, scissors, roads crossing. This chapter collects the angle facts that geometry proofs depend on, especially angles formed when a line crosses two **parallel** lines.

Basic angle pairs

Pair	Property
Complementary	Sum 90°
Supplementary	Sum 180°
Adjacent	Common vertex and arm, non-overlapping
Linear pair	Adjacent and supplementary (form a straight line)
Vertically opposite	Equal

TWO AXIOMS

If a ray stands on a line, the two adjacent angles sum to **180°** (linear pair axiom). Conversely, if two adjacent angles sum to 180°, their outer arms form a line.

Theorem: if two lines intersect, the **vertically opposite angles are equal**.

Parallel lines and a transversal

When a transversal crosses two parallel lines:

- **Corresponding angles** are equal.
- **Alternate interior angles** are equal.
- **Co-interior (consecutive interior) angles** are supplementary (sum 180°).

The **converses** also hold: if any of these is true, the lines are parallel. Lines parallel to the same line are parallel to each other.

AB ∥ CD AND A TRANSVERSAL MAKES ∠1 = 70° WITH AB. FIND THE CO-INTERIOR ANGLE WITH CD.

Co-interior angles sum to 180°, so it is **110°**.

Angle sum property of a triangle

TRIANGLE FACTS

The three angles of a triangle sum to **180°** (proved by drawing a line through one vertex parallel to the opposite side and using alternate angles).
An **exterior angle** equals the **sum of the two interior opposite angles**.

THE ANGLES OF A TRIANGLE ARE IN THE RATIO 2 : 3 : 4. FIND THEM.

$2x + 3x + 4x = 180 \Rightarrow x = 20$. Angles: **40°, 60°, 80°**.

Quick recap

- Linear pair sums to 180°; vertically opposite angles are equal.
- With parallel lines: corresponding equal, alternate equal, co-interior sum 180°.
- Triangle angle sum 180°; exterior angle = sum of interior opposite angles.

Practice questions

1. Two angles form a linear pair. One is 65°. The other is
   (a) 25° (b) 115° (c) 65° (d) 295°
2. Vertically opposite angles are always
   (a) supplementary (b) complementary (c) equal (d) 90°
3. Alternate interior angles formed by a transversal of two parallel lines are
   (a) equal (b) supplementary (c) complementary (d) unequal
4. An exterior angle of a triangle is 110° and one interior opposite angle is 45°. The other interior opposite angle is
   (a) 45° (b) 65° (c) 70° (d) 135°
5. The angles of a triangle are x, 2x and 3x. The largest angle is
   (a) 60° (b) 90° (c) 120° (d) 30°

Answers: 1. b 2. c 3. a 4. b 5. b